

中山大學
SUN YAT-SEN UNIVERSITY

Lecture 1
Introduction

Data Structures and Algorithms

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About me

- B.S. and M.S. in Sun Yat-sen University
- Ph.D. in City University of Hong Kong
- Visiting Scholar at The Hong Kong University of Science and Technology
- Visiting Scholar at New Jersey Institute of Technology
- Visiting Scholar at National University of Singapore
- ACM/ICPC coach
- Research interests
  - Algorithmic design
  - Artificial intelligence
  - Deep reinforcement learning
  - Neural combinatorial optimization

About the Course

- Lectures: 3 class hours per week
- Labs: 2 class hours per week
- Prerequisite courses
  - Programming Design (I)
  - Programming Design (II)
- Successive courses
  - Algorithm Design and Analysis
  - Operating System
  - Artificial Intelligence

- Data Mining

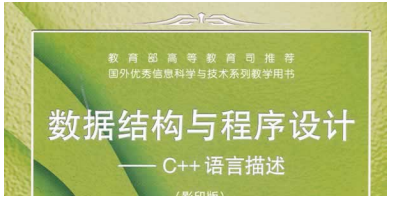
About the Course: Importance

- 专业主干、理论、必修课
- 信息类专业硕士、博士升学考试必考科目
- 后继专业课程的基础课程
- 编程基本技巧与能力训练的关键课程
- 国家软件水平与资格考试必考内容

About the Course: Books

- Textbook:
  - 数据结构（C语言版），严蔚敏，吴伟民，清华大学出版社，第1版，2021年6月
- Reference:
  - Robert L. Kruse and Alexander J. Ryba “Data Structures and Program Design in C++”, Prentice Hall, 1998.
  - 数据结构与算法分析—C++语言描述（第四版）
  - Cormen, Leiserson, Rivest, and Stein. “Introduction to Algorithms”,





McGraw Hill, second edition, 2001.

- D.E. Knuth. “The Art of Computer Programming” (4 volumes in total), Addison-Wesley.
- William Ford and William Topp, “Data Structure with C++ Using STL”, Prentice Hall, second edition, 2002.



About the Course: Platforms

- SYSU platforms
  - https://matrix.sysu.edu.cn/
  - http://soj.acmm.club/
- Commercial platforms
  - https://leetcode.cn/problemset
  - https://ac.nowcoder.com/

- …

About the Course: Grading

- Lecture session
  - 10% Attendance grades
  - 10% Midterm Test
  - 20% Homework & Project
  - 60% Final Exam
- Laboratory session

- 10% Attendance grades
- 60% Reports & Small Tests
- 30% Final Exam

- Every week, you need to sign your names on the attendance book in the lab.

About the Course: Assignments

- Assignments should be submitted to the dedicated locations with names confirmed with required schema and according to formats specified in corresponding specifications.

中山大学计算机学院本科生实验报告

(2024 学年第 1 学期)

课程名称：Data structures and algorithms　　任课教师：张子臻

年级		专业（方向）	
学号		姓名	
电话		Email	
开始日期		完成日期	

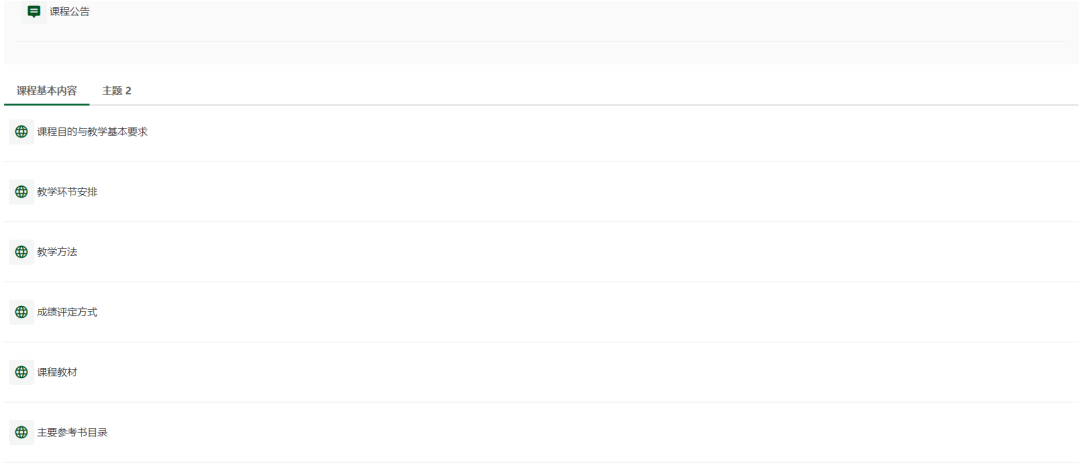
- **NO LATE SUBMISSION!**

About the Course: More Information

Submit your homework via lms.sysu.edu.cn



Join QQ group for discussion


课程公告
课程基本内容
主题 2
课程目的与教学基本要求
教学环节安排
教学方法
成绩评定方式
课程教材
主要参考书目录

Thank you!


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